SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME:CSA1610 **Student Name:-S.Krishna Sagar Reddy**
COURSE CODE: Data Warehousing and Data Mining **REG NO:-192525384**

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05 **Total Marks:50**

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding **(2.5)**	Constraint Analysis **(2.5)**	Solution Development **(2.5)**	Application of Concepts **(1.5)**	Justification **(1)**	**Total(10)**
Weightage	25%	25%	25%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I S.Krishna Sagar Reddy/192525384 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Krishna Sagar Reddy

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID	Symptoms Observed
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:
Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include **"Fever"**. Identify the valid association rules and justify why they satisfy the given constraints.

ANS:-

Given Dataset:-

There are 5 patients, so:

- Minimum support = 50% = at least 3 patients
- Minimum confidence = 70%
- Constraint: Every rule must contain "Fever"

Transactions:

PID	Symptoms
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Step 1: Find frequent itemsets containing Fever

Fever appears in P1, P2, P4, P5:

Support(Fever) = 4/5 = 80% ✓

Fever + Cough appears in P1, P4:

Support = 2/5 = 40% ✗

Fever + Headache appears in P2, P4:

Support = 2/5 = 40% ✗

Fever + Fatigue appears only in P5:

Support = 1/5 = 20% ✗

Therefore, no 2-itemset containing Fever reaches the required 50% support.

Step 2: Check possible rules

A rule such as:

Cough → Fever

has support:

2/5 = 40%, which is below 50%. ✗

Confidence:

$$Confidence(Cough \rightarrow Fever) = \frac{Support(C, oughFever)}{Support(Cough)}$$

Cough appears in P1, P3, P4 = 3 patients.

$$= \frac{2}{3} = 66.67\%$$

It fails both constraints: support < 50% and confidence < 70%.

For:

Headache → Fever

Support = 2/5 = 40% ✗

Confidence:$\frac{2}{3} = 66.67\%$

So it also fails.

Final Answer

There are no valid association rules satisfying all three conditions:

1. Minimum support ≥ 50%
2. Minimum confidence ≥ 70%
3. Rule must include "Fever"

The only frequent itemset containing Fever is {Fever}, with 80% support. However, a rule requires at least two items, and every 2-itemset containing Fever has support below 50%.

Hence, the valid association rule set is: ∅ (no valid rules).

Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:
Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain at least two courses. Explain how the constraint helps in pruning candidate itemsets during mining.

ANS:-

Frequent Itemset Mining with Constraint

There are 5 students.

Minimum support = 40%

Therefore, an itemset must occur in at least:

$$5 \times 40\% = 2$$

So, an itemset is frequent if it appears in at least 2 students.

The constraint is:

Each itemset must contain at least 2 courses.

Step 1: 1-itemsets

Course	Students	Support
Python	4	80%
SQL	3	60%
Data Science	2	40%
Machine Learning	2	40%

All four courses meet the support requirement, but single-course itemsets are removed because of the constraint.

Step 2: 2-itemsets

Itemset	Students	Support
{Python, SQL}	S1, S4 = 2	**40%** ✓
{Python, Data Science}	S2, S4 = 2	**40%** ✓
{Python, Machine Learning}	S5 = 1	20% ✗
{SQL, Data Science}	S4 = 1	20% ✗
{SQL, Machine Learning}	S3 = 1	20% ✗
{Data Science, Machine Learning}	None	0% ✗

Therefore, the frequent 2-itemsets are:

$$\boxed{\{P, ythonSQL\}}$$
$$\boxed{\{P, ythonData\ Science\}}$$

Step 3: 3-itemsets

Possible 3-itemsets include:

- {Python, SQL, Data Science} → appears only in S4 → 20%
- Other 3-itemsets also occur in fewer than 2 students.

Therefore, no 3-itemset satisfies the 40% support requirement.

Final Frequent Itemsets

With minimum support = 40% and at least two courses per itemset:

Frequent Itemset	Support
{Python, SQL}	40%
{Python, Data Science}	40%

How the constraint helps in pruning

The constraint "at least two courses" eliminates all single-course itemsets immediately. Thus, itemsets such as:

- {Python}
- {SQL}
- {Data Science}
- {Machine Learning}

do not need to be considered in the final result.

This reduces the number of candidates that need to be evaluated. After generating 2-itemsets, the minimum support constraint further removes itemsets occurring in fewer than 2 transactions. Therefore, the constraint makes the mining process faster and more efficient by reducing unnecessary candidate itemsets.

Conclusion

The valid frequent itemsets are {Python, SQL} and {Python, Data Science}, both having 40% support and satisfying the requirement of containing at least two courses.

Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:
Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

ANS:-

Apriori Algorithm with Maximum Itemset Size = 3

There are 5 transactions. Since no minimum support is explicitly given, we need one to determine which itemsets are *frequent*. Assuming the same minimum support = 40% as in Q2:

$$5 \times 40\% = 2$$

So, an itemset must appear in at least 2 transactions.

Step 1: 1-Itemsets

Itemset	Occurrences	Support
{Data Mining}	4	80% ✓
{Python Programming}	3	60% ✓
{Database Systems}	3	60% ✓
{Artificial Intelligence}	1	20% ✗

Therefore, the frequent 1-itemsets are:

$$\{Data\ Mining\}, \{Python\ Programming\}, \{Database\ Systems\}$$

Step 2: 2-Itemsets

Itemset	Transactions	Support
{Data Mining, Python Programming}	T1, T4	**40% ✓**
{Data Mining, Database Systems}	T2, T4	**40% ✓**
{Python Programming, Database Systems}	T3, T4	**40% ✓**

All three 2-itemsets are frequent.

Step 3: 3-Itemsets

Possible 3-itemset:

$$\{D, a, ta\ Mining\ Python\ Programming\ Database\ Systems\}$$

It occurs only in T4.

$$Support = \frac{1}{5} \times 100 = 20\%$$

Therefore:

{Data Mining, Python Programming, Database Systems} → 20% ✗

It is not frequent.

Since the constraint says maximum itemset size = 3, we stop after checking 3-itemsets. No itemsets of size 4 or more are generated.

Final Frequent Itemsets

1-itemsets:

- {Data Mining} → 80%
- {Python Programming} → 60%
- {Database Systems} → 60%

2-itemsets:

- {Data Mining, Python Programming} → 40%
- {Data Mining, Database Systems} → 40%
- {Python Programming, Database Systems} → 40%

3-itemsets:

- None

How the Maximum Size Constraint Reduces Complexity

Without the constraint, Apriori could continue generating 4-itemsets, 5-itemsets, etc. as long as enough frequent items existed.

Here, the constraint:

$$\boxed{\text{Maximum itemset size} = 3}$$

means that candidate generation stops after 3-itemsets. Thus:

- No 4-itemset candidates are generated.
- No 5-itemset candidates are generated.
- Fewer support calculations are required.
- Memory usage is reduced.
- The overall Apriori mining process becomes faster.

Conclusion

The maximum-size constraint significantly prunes larger candidate itemsets, reducing the search space and computational cost while ensuring that only itemsets of size 1 to 3 are considered.

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:
Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules.

ANS:-

Multilevel Association Rule Mining

The dataset contains courses organized at different levels of a course hierarchy. We are asked to generate association rules only at the "Computer Science" level.

Step 1: Identify the Computer Science Level

At the required level, we consider:

- Computer Science
- Machine Learning as a course associated with Computer Science

Transactions relevant to this level are:

Student	Courses at Computer Science Level
S1	Computer Science, Machine Learning
S2	—
S3	Computer Science
S4	Computer Science, Machine Learning
S5	—

There are 5 students.

Assuming the same minimum support = 40% used in the previous questions:

$$5 \times 40\% = 2$$

Thus, an itemset must occur in at least 2 students.

Step 2: Calculate Support

Computer Science

Appears in S1, S3, S4:

$$Support = \frac{3}{5} \times 100 = 60\%$$

Machine Learning

Appears in S1, S4:

$$Support = \frac{2}{5} \times 100 = 40\%$$

Computer Science + Machine Learning

Appears in S1 and S4:

$$Support = \frac{2}{5} \times 100 = 40\%$$

Therefore, the frequent itemset is:

$$\boxed{\{C, omputer\ Science\ Machine\ Learning\}}$$

Step 3: Association Rule

The possible rule is:

$$Computer\ Science \rightarrow Machine\ Learning$$

Confidence:

$$Confidence = \frac{Support(C, omputer\ ScienceMachine\ Learning)}{Support(Computer\ Science)}$$

$$= \frac{2}{3} = 66.67\%$$

The reverse rule is:

$$Machine\ Learning \rightarrow Computer\ Science$$

Confidence:

$$= \frac{2}{2} = 100\%$$

Therefore:

Association Rule	**Support**	**Confidence**
Computer Science → Machine Learning	40%	66.67%
Machine Learning → Computer Science	40%	**100%**

If the minimum confidence is **70%**, only:

$$\boxed{Machine\ Learning \rightarrow Computer\ Science}$$

is a valid rule.

Effect of Course Hierarchy:-

A course hierarchy allows association mining to be performed at different levels of detail.

For example:

Engineering → Computer Science → Machine Learning

At a higher level, we may discover broad relationships involving Engineering. At the Computer Science level, the mining becomes more specific and reveals the relationship between Computer Science and Machine Learning.

The constraint "only Computer Science level" prevents rules involving unrelated higher- or lower-level categories from being generated. This reduces the search space and produces more focused and meaningful association rules.

Conclusion

At the Computer Science level, the important relationship is:

$$\boxed{Machine\ Learning \rightarrow Computer\ Science}$$

with 40% support and 100% confidence (assuming minimum confidence = 70%). The hierarchy helps control the level of abstraction and reduces unnecessary candidate rules.

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:
Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

ANS:-

Multidimensional Association Rule Mining

The constraint is:

Age Group = Young

So, we first select only customers whose age group is Young.

Step 1: Filter the Dataset

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C3	Young	Low	Travel Insurance
C5	Young	Medium	Vehicle Insurance

Thus, there are 3 Young customers.

Step 2: Identify Patterns

Pattern 1: Young + High Income → Health Insurance

C1 has:

- Age Group = Young
- Income = High
- Policy = Health Insurance

This shows that high-income young customers in this small dataset are associated with Health Insurance.

Pattern 2: Young + Low Income → Travel Insurance

C3 has:

- Age Group = Young
- Income = Low
- Policy = Travel Insurance

This suggests an association between low-income young customers and Travel Insurance.

Pattern 3: Young + Medium Income → Vehicle Insurance

C5 has:

- Age Group = Young
- Income = Medium
- Policy = Vehicle Insurance

This indicates an association between medium-income young customers and Vehicle Insurance.

Summary

Constraint	Income Level	Policy	Observation
Young	High	Health Insurance	High-income young customer
Young	Low	Travel Insurance	Low-income young customer
Young	Medium	Vehicle Insurance	Medium-income young customer

Because there is only one customer in each income category, these are observed patterns rather than statistically strong association rules. More customer records would be needed to establish reliable support and confidence.

How It Supports Targeted Marketing:-

The patterns can help an insurance company divide young customers into different marketing segments:

- Young + High Income: Promote comprehensive Health Insurance plans with additional benefits.
- Young + Medium Income: Promote affordable Vehicle Insurance packages.
- Young + Low Income: Offer budget-friendly Travel Insurance plans or flexible payment options.

This is an example of multidimensional association mining because multiple attributes—Age Group, Income Level, and Policy Purchased—are analyzed together.

Conclusion

By applying the constraint Age Group = Young, the analysis focuses only on the target customer segment. The dataset suggests three income-policy patterns: High → Health Insurance, Medium → Vehicle Insurance, and Low → Travel Insurance. These patterns can help businesses design more personalized marketing campaigns and recommend suitable insurance products.